

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET C — NCERT EXEMPLAR, IN-TEXT SOURCES & TRICKY QUESTIONS****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-C-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET C - NCERT Exemplar & Tricky Questions):

1. The question paper comprises **25 NCERT Exemplar & Critical Application Questions** of **1 mark each**.
2. Tests **in-text source boxes, fine glossary definitions, historical source extracts, and tricky boundary cases**.
3. All questions are compulsory. There is no negative marking for incorrect answers.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

**NCERT IN-TEXT & EXEMPLAR TIER:** Questions in this set explore subtle details, specialized definitions, and non-intuitive concepts drawn directly from official NCERT text margins, source boxes, and glossary entries.**Q.1** If the sea-level temperature at Mumbai is 30 °C, what is the theoretical air temperature atop Mahabaleshwar (elevation ~1,400 meters) assuming a normal environmental lapse rate of 6.5 °C per km?

- (A) 20.9 °C (B) 25.5 °C
(C) 15.0 °C (D) 35.2 °C

Q.2 Why is the Coriolis force zero at the equator and maximum at the geographic poles?

- (A) The horizontal component of the Coriolis acceleration is proportional to the sine of the latitude ($\sin \phi$), and $\sin(0^\circ) = 0$ while $\sin(90^\circ) = 1$
(B) Earth does not rotate at the equator and spins only at the poles
(C) Gravity is completely absent at the equator
(D) Equatorial air molecules are twice as heavy as polar air molecules

Q.3 What is the meteorological distinction between 'Mango Showers' in Kerala and 'Kalbaisakhi' in West Bengal?

- (A) Mango Showers are gentle pre-monsoon convective rains aiding mango ripening; Kalbaisakhi (Norwesters) are violent, destructive thunderstorms with squalls and torrential rain originating in the Chota Nagpur plateau
(B) Mango Showers drop ripe mango fruits from trees; Kalbaisakhi is a winter snowfall
(C) Kalbaisakhi occurs in Rajasthan; Mango Showers occur in the Himalayas
(D) Both names refer to identical quiet fog that occurs in December

Q.4 Why is the global warming potential (GWP) of Methane (CH_4) rated roughly 28–36 times higher than that of Carbon Dioxide (CO_2) over a 100-year timescale?

- (A) Methane is far more effective at absorbing outgoing infrared radiation per molecule than CO_2 , despite having a shorter atmospheric lifetime
(B) Methane explodes into flames whenever sunlight touches it in clouds
(C) Methane is a liquid that falls as boiling acid rain
(D) Carbon dioxide has zero greenhouse warming potential

Q.5 Why do coastal regions experience an onshore 'Sea Breeze' during daytime and an offshore 'Land Breeze' during night hours?

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| <p>(A) Daytime solar heating warms land faster than water, creating low pressure over land that pulls cool air from the sea; at night, land cools faster than water, reversing the thermal pressure gradient</p> | <p>(B) The ocean breathes in during daytime and breathes out during night hours</p> |
| <p>(C) Daytime winds are pushed by marine cargo ships into coastal ports</p> | <p>(D) The moon's magnetic field repels daytime winds from the sea</p> |

Q.6 What is the 'Horse Latitudes' in global atmospheric circulation, and why were they given that historical name?

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| <p>(A) Subtropical high-pressure belts (~30°–35° N and S) with calm, subsiding dry air, where Spanish sailing ships were often becalmed and threw horses overboard to conserve water</p> | <p>(B) Equatorial plains where wild horses were bred for cavalry units</p> |
| <p>(C) Polar regions where horses hibernate during the winter months</p> | <p>(D) A constellation of stars that resembles a galloping horse</p> |

Q.7 What is the 'Tropical Easterly Jet Stream' and what is its specific connection to the Indian summer monsoon?

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| <p>(A) An upper-tropospheric easterly wind stream (~14°N) driven by the intense thermal heating of the Tibetan Plateau, which steers southwest monsoon moisture into peninsular India</p> | <p>(B) A high-speed train running across the northern plains of India</p> |
| <p>(C) A polar jet stream that causes blizzards across southern India</p> | <p>(D) A surface sea current that flows backwards across the Arabian Sea</p> |

Q.8 Why does the leeward side of a mountain range experience 'adiabatic warming' as descending air moves down the slope?

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| <p>(A) Increasing atmospheric pressure at lower elevations compresses the descending air parcel, raising its temperature at the dry adiabatic lapse rate (~10 °C per km)</p> | <p>(B) The air parcel absorbs thermal heat from underground volcanic rocks</p> |
| <p>(C) Descending air friction with trees causes air molecules to ignite</p> | <p>(D) The mountain acts as a giant solar radiator that points only downward</p> |

Q.9 What is the 'Southern Oscillation Index' (SOI) mathematically based upon?

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| <p>(A) The normalized sea-level atmospheric pressure difference between Tahiti (central Pacific) and Darwin (northern Australia)</p> | <p>(B) The height of tidal waves measured at the South Pole</p> |
| <p>(C) The number of earthquakes recorded along the San Andreas Fault</p> | <p>(D) The temperature difference between New York and London</p> |

Q.10 Why does the atmosphere contain roughly 21% Oxygen (\$O_2\$) and 78% Nitrogen (\$N_2\$), but Carbon Dioxide (\$CO_2\$) is only 0.04% (420 ppm)?

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| <p>(A) Despite its tiny fractional concentration, \$CO_2\$ is a potent dipolar greenhouse gas that exerts massive radiative forcing, whereas diatomic \$N_2\$ and \$O_2\$ cannot absorb infrared heat</p> | <p>(B) Nitrogen and oxygen are toxic gases that heat the planet violently</p> |
| <p>(C) Carbon dioxide was added to the air exclusively in the year 2000</p> | <p>(D) Oxygen absorbs all ultraviolet radiation from the sun</p> |

Q.11 What causes the 'Breaks' (dry spells) in the Indian southwest monsoon during July and August?

- (A) The seasonal monsoon trough shifts northward to the Himalayan foothills, causing rain to cease over the fertile northern plains while triggering floods in Himalayan river valleys
- (B) The sun disappears from the sky for two weeks in July
- (C) The Indian ocean runs out of water to evaporate
- (D) The trade winds turn into solid ice over central India

Q.12 Why is the Mesosphere the coldest layer of the atmosphere (temperatures dropping to -90 °C at the mesopause)?

- (A) It lacks ozone to absorb UV radiation, has virtually no water vapor to trap infrared, and cools efficiently through CO_2 radiative emissions into space
- (B) It is surrounded on all sides by frozen icebergs
- (C) The mesosphere is located beyond the orbit of the planet Mars
- (D) Meteors drop chunks of dry ice into the mesosphere daily

Q.13 What is the 'Urban Heat Island' (UHI) effect observed in major metropolitan cities like Delhi and Mumbai?

- (A) Concrete buildings, asphalt roads, vehicular heat, and reduced vegetative evapotranspiration cause urban centers to remain 2–6 °C warmer than surrounding rural hinterlands
- (B) Cities are built directly on top of active volcanic magma chambers
- (C) Rural villages use giant air-conditioners to freeze their farmland
- (D) The sun only shines on cities and completely avoids rural villages

Q.14 What is the 'dew point' temperature in atmospheric humidity measurements?

- (A) The temperature to which an air parcel must be cooled at constant pressure to reach 100% relative humidity (water vapor saturation), initiating condensation
- (B) The boiling point of liquid water at standard sea level pressure
- (C) The temperature at which raindrops freeze into solid hail blocks
- (D) The average temperature of the Indian Ocean during winter

Q.15 Why does high-altitude cirrus cloud cover have a net warming effect on Earth, whereas low-altitude thick stratocumulus clouds have a net cooling effect?

- (A) Cirrus clouds are thin and allow shortwave solar light to pass while trapping outgoing terrestrial infrared; low thick clouds have high albedo that reflects solar light back to space
- (B) Cirrus clouds are filled with burning fire; stratocumulus clouds are filled with snow
- (C) Stratocumulus clouds absorb all atmospheric nitrogen
- (D) Both cloud types exert identical 100% warming across the globe

Q.16 What is the meteorological cause of the 'Loo' winds blowing across northwestern India in May and June?

- (A) Intense sensible heating of the arid Thar Desert and Balochistan plateaus generates scorching, dry, low-density thermal winds blowing down the pressure gradient across the Ganga plain
- (B) A sudden cooling of the Himalayan peaks in mid-summer
- (C) Evaporation of the Caspian Sea in Central Asia
- (D) Marine breezes blowing directly from the Bay of Bengal

Q.17 Why does the sky appear deep blue on a clear sunny day according to Rayleigh scattering?

- (A) Atmospheric gas molecules are much smaller than the wavelength of light and scatter short-wavelength blue light ($\sim \lambda^{-4}$) far more efficiently than red light
- (B) The ocean reflects its blue water directly onto the clouds
- (C) Oxygen gas is naturally a bright blue liquid in the air
- (D) Outer space is filled with blue neon fluorescent lighting

Q.18 What is the environmental hazard caused by 'acid rain' ($\text{pH} < 5.6$) in industrialized regions?

- (A) Emissions of sulfur dioxide (SO_2) and nitrogen oxides (NO_x) react with water vapor to form sulfuric and nitric acids, which acidify lakes, kill fish, and corrode limestone monuments
- (B) Acid rain causes oceans to freeze into solid dry ice
- (C) Acid rain turns soil into pure metallic copper
- (D) Acid rain causes the atmosphere to run out of oxygen

Q.19 How does the 'Polar Vortex' in winter maintain extremely cold temperatures over the Arctic?

- (A) A persistent, large-scale low-pressure system and strong circumpolar jet stream keep freezing Arctic air trapped within the polar basin; when it weakens, cold air plunges south
- (B) The North Pole is cooled by giant mechanical fans
- (C) The earth's rotation stops completely in the Arctic during winter
- (D) The Arctic ocean is made of liquid liquid helium

Q.20 What is 'radiosonde' technology used daily by national weather meteorological agencies?

- (A) A battery-powered telemetry instrument package carried into the stratosphere by a weather balloon to transmit pressure, temperature, humidity, and wind velocity profiles
- (B) An underground microphone that listens to earthquake sounds
- (C) A camera that takes photographs of deep ocean fish
- (D) A satellite dish installed on the roof of high school buildings

Q.21 Why does Tamil Nadu receive copious rainfall from the 'Northeast Monsoon' (Retreating Monsoon) in October, November, and December?

- (A) Winds blowing from the high-pressure landmass of northern India cross the warm Bay of Bengal, gather moisture, and strike the Coromandel Coast as onshore rainy winds
- (B) Tamil Nadu is located in the Arctic circle during winter
- (C) The Arabian Sea reverses its water into the Bay of Bengal
- (D) Himalayan glaciers melt directly into Tamil Nadu rivers

Q.22 What is the 'Intergovernmental Panel on Climate Change' (IPCC)?

- (A) A United Nations scientific body that assesses the state of knowledge on climate change, its environmental and socio-economic impacts, and potential mitigation response options
- (B) An international political party that contests elections across Europe
- (C) A corporate company that manufactures oil drilling rigs
- (D) A military defense alliance that operates naval fleets

Q.23 What does a barometer reading that 'drops rapidly' within a few hours signify to a meteorologist?

- (A) The rapid approach of a severe low-pressure storm system or cyclone, accompanied by violent winds and heavy precipitation
- (B) The arrival of calm, dry, cloudless sunny weather for the next week
- (C) That the barometer battery has completely died
- (D) That atmospheric air has ceased to have any weight

Q.24 Why do polar bears in the Arctic suffer when global warming accelerates summer sea-ice loss?

- (A) They rely on sea-ice platforms to hunt seals; shrinking ice limits their hunting range, forcing them into prolonged starvation and reproductive decline
- (B) They prefer to swim in boiling water rather than cold water
- (C) Polar bears eat only tropical mangoes and bananas
- (D) Global warming freezes their fur into solid ice

Q.25 In summary, what is the ultimate driver of all atmospheric winds, weather phenomena, and ocean currents on planet Earth?

- (A) The unequal solar heating of the Earth between the warm equator and cold poles, and Earth's ongoing effort to redistribute thermal heat via convection and rotation
- (B) The rotation of the moon around the earth
- (C) Underground volcanic magma boiling the oceans
- (D) The burning of fossil fuels by human cars

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET C)

Q. No.	Correct	Curricular Reference & In-Depth Conceptual Solution
Q.1	(A)	Temperature decrease $= 1.4 \text{ km} \times 6.5^\circ \text{C/km} = 9.1^\circ \text{C}$. Temperature at Mahabaleshwar $= 30^\circ \text{C} - 9.1^\circ \text{C} = 20.9^\circ \text{C}$.
Q.2	(A)	Coriolis parameter $f = 2\Omega \sin \phi$. At the equator ($\phi = 0^\circ$), $f = 0$ (zero deflection); at the poles ($\phi = 90^\circ$), $\sin 90^\circ = 1$ (maximum deflection).
Q.3	(A)	Kalbaisakhi ('calamity of Baisakh') brings violent squalls (up to 100 km/h) and hail to Bengal/Assam, while southern pre-monsoon showers are milder and aid coffee/mango crops.
Q.4	(A)	Methane's radiative efficiency is far higher because its fundamental vibrational modes absorb infrared in window bands where CO_2 is already saturated.
Q.5	(A)	Differential specific heat: land heats and cools rapidly ($c_{\text{land}} \ll c_{\text{water}}$), establishing diurnal pressure reversals (sea breeze by day, land breeze by night).
Q.6	(A)	Subtropical Highs feature descending air (subsidence) and light variable winds; stranded Spanish caravels historically jettisoned horses to conserve drinking water.
Q.7	(A)	Intense summer heating of the Tibetan Plateau generates the upper-air Tropical Easterly Jet (TEJ), whose descending branch over the Arabian Sea drives the monsoon low-level jet.
Q.8	(A)	As air descends into higher pressure, work is done on the parcel by compression, raising its internal thermal energy and temperature at the Dry Adiabatic Lapse Rate (DALR, $\sim 9.8^\circ \text{C/km}$).
Q.9	(A)	SOI $= P_{\text{Tahiti}} - P_{\text{Darwin}}$. Sustained negative SOI values indicate El Niño conditions (warm Pacific, weak Walker cell), often causing deficient monsoon rain in India.
Q.10	(A)	Symmetric homonuclear diatomic molecules (N_2 , O_2) have no dipole moment and cannot absorb infrared; triatomic CO_2 has bending modes that make it a powerful greenhouse gas.
Q.11	(A)	When the monsoon trough shifts north against the Himalayas, plains experience dry spells ('breaks'), while heavy orographic downpours in the mountains trigger devastating river floods.
Q.12	(A)	The mesosphere is far above the stratospheric ozone layer and has minimal density, radiating heat into space via CO_2 without absorbing solar radiation, cooling to $\sim -90^\circ \text{C}$.
Q.13	(A)	High thermal mass of concrete/asphalt, reduced latent heat flux from vegetation loss, and anthropogenic waste heat create elevated urban temperatures (UHI effect).
Q.14	(A)	The dew point is the saturation temperature where actual vapor pressure equals saturation vapor pressure ($e = e_s$); cooling below this temperature produces dew, fog, or rain.
Q.15	(A)	Thin cirrus clouds are transparent to solar insolation but opaque to terrestrial infrared (greenhouse warming); thick low stratocumulus clouds reflect 70–80% of sunlight (albedo cooling).
Q.16	(A)	Intense solar heating over the Thar Desert creates extreme thermal low pressure, driving dry, blistering winds (temperatures $45^\circ\text{--}48^\circ \text{C}$) eastward across Punjab, Haryana, and UP.
Q.17	(A)	Rayleigh scattering intensity is inversely proportional to the fourth power of wavelength ($I \propto 1/\lambda^4$); blue light ($\sim 400 \text{ nm}$) scatters ~ 10 times more than red light ($\sim 700 \text{ nm}$).
Q.18	(A)	Atmospheric oxidation of SO_2 and NO_x produces H_2SO_4 and HNO_3 , dropping precipitation pH below 5.6, leaching soil nutrients, killing aquatic life, and eroding marble (e.g., Taj Mahal).
Q.19	(A)	A tight cyclonic circumpolar jet stream confines frigid stratospheric air; sudden stratospheric warmings (SSWs) disrupt the vortex, undulating the jet and dumping Arctic air into mid-latitudes.

Q. No.	Correct	Curricular Reference & In-Depth Conceptual Solution
Q.20	(A)	Weather balloons launch radiosonde packages twice daily worldwide, capturing vertical atmospheric profiles (soundings) essential for numerical weather prediction models.
Q.21	(A)	Retreating continental winds traverse the Bay of Bengal, picking up moisture; upon reaching the Coromandel Coast, orographic and coastal convergence generates heavy winter rains.
Q.22	(A)	Established by UNEP and WMO in 1988, the Nobel-winning IPCC synthesizes peer-reviewed climate science into landmark Assessment Reports (AR6) for global policymakers.
Q.23	(A)	A sharp barometric pressure drop indicates steep pressure gradient forces converging around an approaching deep cyclonic depression, signaling impending severe weather.
Q.24	(A)	Sea ice is an obligatory foraging platform for polar bears (*Ursus maritimus*) to access ringed seals; premature ice breakup shortens hunting seasons, depressing body mass.
Q.25	(A)	Differential solar insolation (net radiative surplus at the equator, deficit at the poles) powers Earth's giant thermodynamic heat engine, driving winds and ocean gyres.